

Assignment No - 04 (Group A)

Problem statement:

Data Analytics I

Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (<https://www.kaggle.com/c/boston-housing>). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.

The objective is to predict the value of prices of the house using the given features.

Pre-requisite

1. Basic of Python Programming
2. Concept of Linear Regression

Objective

Students should be able to perform the Statistical operations using Python on any open source dataset.

Software and Hardware requirements:-

1. **Operating system:** Linux- Ubuntu 16.04 to 17.10, or Windows 7 to 10,
2. **RAM-** 2GB RAM (4GB preferable)
3. **IDE :-** Anaconda Jupiter Notebook / pycharm / Visual Studio

Theory-

1. Linear Regression:

It is a machine learning algorithm based on supervised learning. It targets prediction values on the basis of independent variables.

It is preferred to find out the relationship between forecasting and variables. A linear relationship between a dependent variable (X) is continuous; while independent variable(Y) relationship may be continuous or discrete. A linear relationship should be available in between predictor and target variable so known as Linear Regression.

Linear regression is popular because the cost function is Mean Squared Error (MSE) which is equal to the average squared difference between an observation's actual and predicted values.

It is shown as an equation of line like :

$$Y = m \cdot X + b + e$$

Where : b is intercepted, m is slope of the line and e is error term.

This equation can be used to predict the value of target variable Y based on given predictor variable(s) X , as shown in Fig. 1.

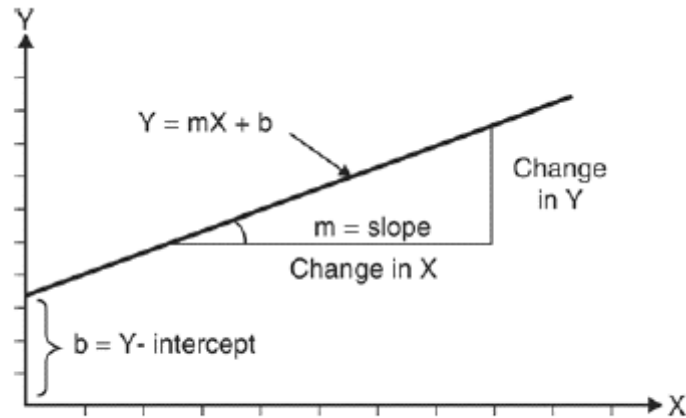


Fig. 1: geometry of linear regression

Fig. 2 shown below is about the relation between weight (in Kg) and height (in cm), a linear relation. It is an approach of studying in a statistical manner to summarise and learn the relationships among continuous (quantitative) variables.

Here a variable, denoted by 'x' is considered as the predictor, explanatory, or independent variable.

Another variable, denoted 'y', is considered as the response, outcome, or dependent variable. While "predictor" and "response" used to refer to these variables.

Simple linear regression technique concerned with the study of only one predictor variable.

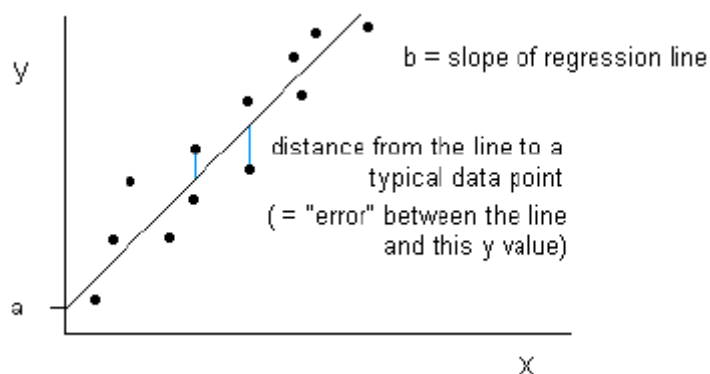


Fig.2 : Relation between weight (in Kg) and height (in cm)

Multi-Variate Regression:

It concerns the study of two or more predictor variables.

Usually a transformation of the original features into polynomial features from a given degree is preferred and further Linear Regression is applied on it.

A simple linear model $Y = a + bX$ in original feature will be transformed into polynomial feature is transformed and further a linear regression applied to it and it will be something like

$$Y = a + bX + cX^2$$

If a high degree value is used in transformation the curve becomes over-fitted as it captures the noise from data as well.

2. Least Square Method for Linear Regression

Linear Regression involves establishing linear relationships between dependent and independent variables. Such a relationship is portrayed in the form of an equation also known as the linear model.

A simple linear model is the one which involves only one dependent and one independent variable. Regression Models are usually denoted in Matrix Notations.

However, for a simple univariate linear model, it can be denoted by the regression equation

$$\hat{y} = \beta_0 + \beta_1 x$$

where $\hat{y}$ is the dependent or the response variable

x is the independent or the input variable

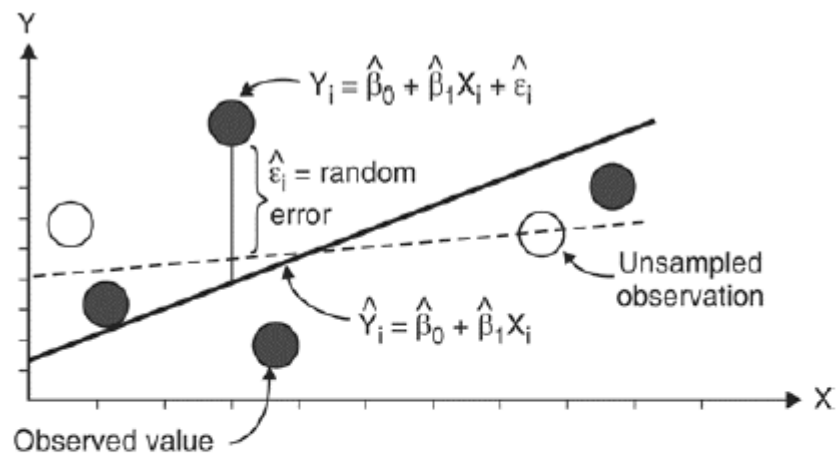
β_0 is the value of y when $x=0$ or the y intercept

β_1 is the value of slope of the line ϵ is the error or the noise

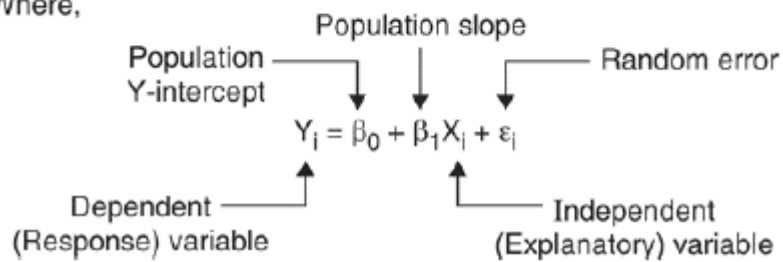
This linear equation represents a line also known as the 'regression line'. The least square estimation technique is one of the basic techniques used to guess the values of the parameters and based on a sample set.

This technique estimates parameters β_0 and β_1 by trying to minimise the square of errors at all the points in the sample set. The error is the deviation of the actual sample data point from the regression line. The technique can be represented by the equation.

$$\min \sum_{i=0}^n (\hat{y}_i - y_i)^2 \quad (2)$$



Where,



Using differential calculus on equation 1 we can find the values of β and such $0 \leq \beta \leq 1$ that the sum of squares (that is equation 2) is minimum

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (3)$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} \quad (4)$$

Once the Linear Model is estimated using equations (3) and (4), we can estimate the value of the dependent variable in the given range only. Going outside the range is called extrapolation which is inaccurate if simple regression techniques are used.

3. Measuring Performance of Linear Regression

Mean Square Error:

The Mean squared error (MSE) represents the error of the estimator or predictive model created based on the given set of observations in the sample. Two or more regression models created using a given sample data can be compared based on their MSE. The lesser the MSE, the better the regression model is. When the linear regression model is trained using a given set of observations, the model with the least mean sum of squares error (MSE) is selected as the best model. The Python or R packages select the best-fit model as the model with the lowest MSE or lowest RMSE when training the linear regression models.

Mathematically, the MSE can be calculated as the average sum of the squared difference between the actual value and the predicted or estimated value represented by the regression model (line or plane).

$$MSE = \frac{1}{n} \sum \underbrace{\left(y - \hat{y} \right)^2}_{\substack{\text{The square of the difference} \\ \text{between actual and} \\ \text{predicted}}}$$

An MSE of zero (0) represents the fact that the predictor is a perfect predictor.

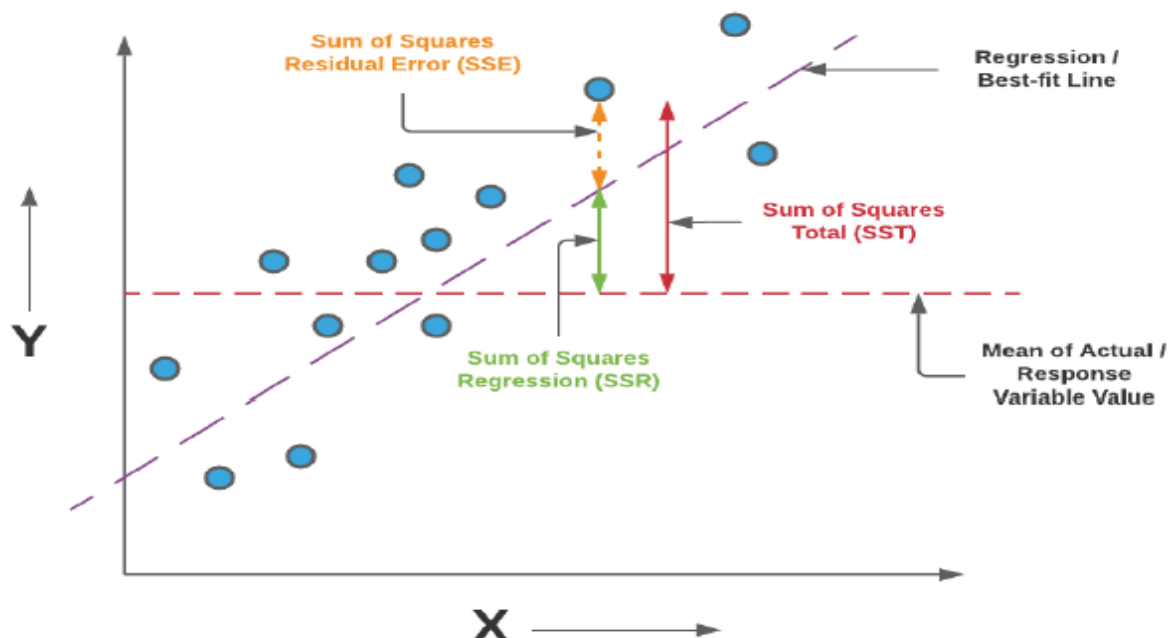
RMSE:

Root Mean Squared Error method that basically calculates the least-squares error and takes a root of the summed values.

Mathematically speaking, Root Mean Squared Error is the square root of the sum of all errors divided by the total number of values. This is the formula to calculate RMSE

$$RMSE = \sqrt{\sum_{i=1}^n \frac{1}{n} (\hat{y}_i - y_i)^2}$$

RMSE - Least Squares Regression Method – Edureka R-Squared :



R-Squared is the ratio of the sum of squares regression (SSR) and the sum of squares total (SST).

SST : total sum of squares (SST), regression sum of squares (SSR), Sum of square of errors (SSE) are all showing the variation with different measures.

$$SST = \sum_{i=1} (y_i - \bar{y})^2$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$R^2 = \frac{SSR}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

Conclusion:

In this way we have done data analysis using linear regression for Boston Dataset and predict the price of houses using the features of the Boston Dataset.